

Unstructured Primal-Dual Mesh Improvement and Generation



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Goal: good primal (triangle) properties and dual (Voronoi) properties *simultaneously*
Discrete Exterior Calculus (DEC) PDE simulations

Do you have a DEC simulation?

We want to partner, discover what makes a difference in practice!

- Mathematical advances

Analyzed metric bounding Hodge-star discretization error
Designed new metric with good optimization behavior

Features of $\overline{\text{HOT}}$:

- $\overline{\text{HOT}}(\mathcal{M}(x, y)) \rightarrow \infty$ as $(x, y) \rightarrow \mathbf{P}$
(has inversion barrier)
- $\overline{\text{HOT}}$ is dimensionless
- $\overline{\text{HOT}}$ is scale invariant

$$\text{HOT}(\sigma) = |\sigma| * \sigma |W(\mu_\sigma, \mu_{*\sigma})|^2 = \frac{\mathbf{a}^3 |\sigma|}{3} + \frac{\mathbf{a} |\sigma|^3}{12} + \frac{\mathbf{b}^3 |\sigma|}{3} + \frac{\mathbf{b} |\sigma|^3}{12}$$

$$\overline{\text{HOT}}(\sigma) = \frac{1}{\text{area}(\triangle)^2} \left(\frac{\mathbf{a}^3 |\sigma|}{3} + \frac{\mathbf{a} |\sigma|^3}{12} \right) + \frac{1}{\text{area}(\triangle)^2} \left(\frac{\mathbf{b}^3 |\sigma|}{3} + \frac{\mathbf{b} |\sigma|^3}{12} \right)$$

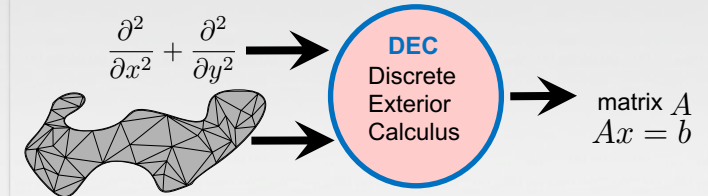
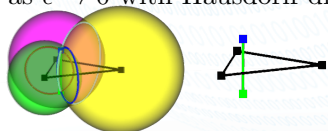
- Mesh-improvement software framework
resampling, constrained to improve quality
traditional smoothing, insertion/deletion

Goal: dual Voronoi polyhedral meshing, VoroCrust

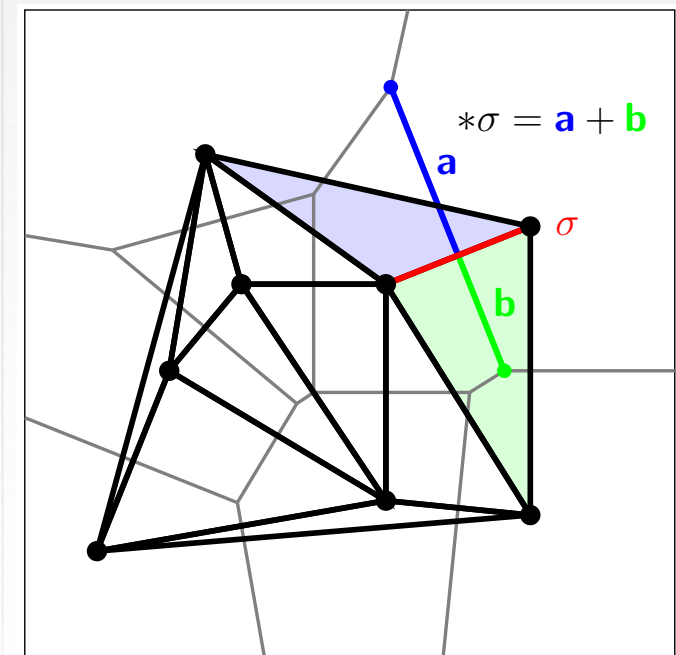
Do you need polyhedral meshes? Tell us!

- Math advantages of boundary-conforming Voronoi cells
flat facets, convex cells, aspect ratio bounds
- Derived sufficient conditions for provably correct meshes

- Theorem (Sampling Conditions are Sufficient.)** VoroCrust produces a geometrically-close and topologically-correct reconstruction when the sampling conditions are satisfied. Moreover, with $\delta = 8\epsilon \leq 1/20$, mesh $\hat{\mathcal{M}} \rightarrow \mathcal{M}$ manifold as $\epsilon \rightarrow 0$ with Hausdorff distance $h < 2 \delta \sin(\theta)$ lfs $< 5.0 \delta^2$.



Discretization error < HOT energy(primal-dual $\sigma * \sigma$)



References

- Meshes Optimized for Discrete Exterior Calculus, tech report
- A Constrained Resampling Strategy for Mesh Improvement, CGF
- Disk Density Tuning of a Maximal Random Packing, CGF
- Spoke-Darts for High-Dimensional Blue Noise Sampling, accepted to TOG
- A Seed Placement Strategy for Conforming Voronoi Meshing, CCCG
- VoroCrust Surface Reconstruction Conditions, manuscript
- VoroCrust: Clipping-Free Voronoi Meshing, manuscript

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